Last name ______________________

First name ______________________

LARSON—MATH 310–CLASSROOM WORKSHEET 05
Distance, Triangle Inequality, Standard Deviation.

Review

1. How can we write the linear function $f(x, y) = 2x + 3y$ as an inner product?
2. Why do all linear functions go through the origin?
3. Why is the function $f(x, y) = 2x + 3y + 5$ *not* linear?
4. Why is the function $f(x, y) = 2x + 3y + 5$ an *affine* function?

Norm and Distance (Chp. 3)

The **norm** of a vector $\vec{x} = (x_1, x_2, ..., x_n)$ is $||\vec{x}|| = \sqrt{x_1^2 + x_2^2 + \ldots x_n^2}$.

1. What is the *norm* of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$?

The **root-mean square** of a n-vector $\vec{v}$ is $rms(\vec{v}) = \frac{||\vec{v}||}{\sqrt{n}}$.

2. Find $rms(\vec{v})$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.

New (from Chp.3)

3. Find $dist(\vec{a}, \vec{b})$ where $\vec{a} = (1, 2, 3)$ and $\vec{b} = (1, 0, -7)$.

4. Find $\vec{1}^T\vec{v}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.

5. Find $\vec{v}^T\vec{1}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.

6. Find $avg(\vec{v})$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.

7. Find the *de-meaned* vector $\tilde{v}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.

Find the *standard deviation* $std(\vec{v})$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.